Kelp-3 Buoy Field Trial Report 2025

Brightwater Labs - Hardware Engineering. Fictional document for testing.

Executive Summary

The Kelp-3 buoy was trialled for 120 days between April and July 2025 at four sites in the North Sea and the Baltic. The trial met its main goal: average uptime was 99.1 percent, above the 98 percent target. The main weakness was the anchor chain swivel, which failed on two buoys.

We recommend a production run of 200 units in 2026, conditional on a redesigned swivel.

1. Trial Setup

Twelve Kelp-3 buoys were deployed: four at Helgoland, three at Fehmarn, three at Rugen and two at Bornholm. Each buoy carried a wave sensor, a water-temperature string with 8 sensors down to 20 m, and a solar charging system with a 240 Wh lithium iron phosphate battery.

Data was transmitted every 10 minutes over a satellite link. When the link was unavailable, readings were buffered on board for up to 14 days.

2. Power and Battery Results

The solar panels produced on average 410 Wh per day in May and 520 Wh per day in June. The lowest daily yield, 95 Wh, was recorded at Bornholm during a storm on 14 April.

Battery state of charge never dropped below 38 percent. Battery capacity loss after 120 days was 1.6 percent, which projects to a battery service life of about seven years.

3. Data Quality

Wave height readings were compared against a reference Waverider buoy at Helgoland. The mean absolute difference was 0.07 m. Water-temperature sensors drifted by less than 0.05 degrees Celsius.

Biofouling on the temperature string became visible after 60 days. An anti-fouling copper sleeve is recommended for deployments longer than three months.

4. Failures and Incidents

Two anchor chain swivels failed due to corrosion, at Fehmarn on day 71 and at Rugen on day 93. Both buoys drifted but were recovered within 36 hours thanks to the GPS tracker.

One satellite modem froze and needed a remote power cycle. Firmware version 2.4.1 fixed the cause, a watchdog timer that was disabled by mistake.

5. Costs and Recommendations

The bill of materials per buoy was 4,850 euros, 6 percent above the estimate, mainly because of the satellite modem price. Deployment cost per buoy including vessel time was 1,900 euros.

Recommendations: (1) replace the galvanised swivel with a duplex stainless steel swivel; (2) add the copper anti-fouling sleeve; (3) ship firmware 2.4.1 or newer; (4) proceed with a production run of 200 units in 2026.